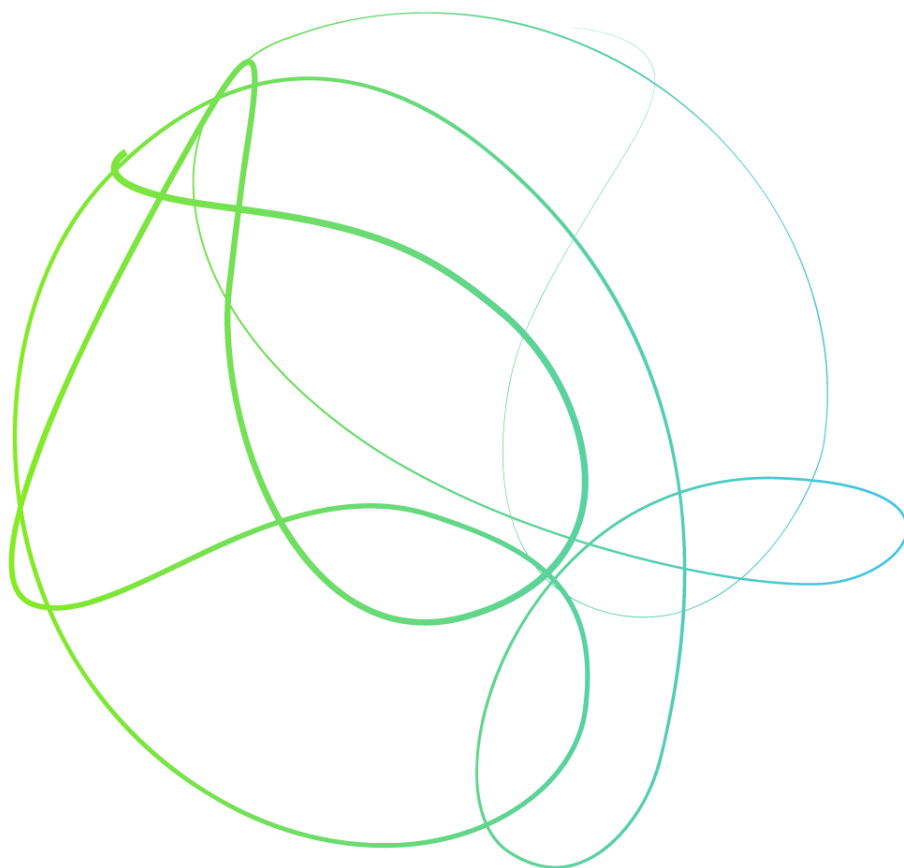




Issue Tracking System Using React JS

Version Dated: 06.2026

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Problem Statement

Objective

Stark Industries, a giant conglomerate led by Tony Stark has collaborated with Deloitte USI to build a web application to allow customers to configure an issue tracking system. As the instance is built, the application should display in what stage the issue is.

Business Context

Mr. Stark, apart from being a genius, billionaire, and philanthropist, is also a businessman who would like to create this Issue Tracking System, helping the stakeholders have a real-time online portal to track issues faced in the business. Mr. Stark would like a rich, interactive, and pleasing experience with solid functionality. Tony loves young talent and hence has come up to Deloitte USI for an optimized solution.

Project Plan

Mr. Stark has an official event scheduled on Friday at 12:30 pm, where he wants to launch the Issue Tracker System. The delivery is expected to be of high quality, iterative, and showing progressive enhancements without waiting for a last-minute integration. This will reduce the risk and keep Mr. Stark informed about the progress schedule instead of a last-minute surprise. There are daily checkpoints twice a day to keep track of the progress.

The technical team has estimated the project to be doable by an experienced engineer in approximately 8 hrs. Given that the Engineering team is newly introduced to React, they are expected to take slightly longer. It's expected that they will heavily prioritize doing the important stuff to make the presentation on Friday successful.

The milestones below show the must-have features and how they should be planned for delivery. The remaining time should be used to complete the functionality and prioritized based on the business context.

QUALITY FOCUS

The deliverable must adhere to all the industry standards in terms of functional and technical aspects creating a high value asset that helps the users to rely on the system going forward for any project-related information as far as the issues faced with the business are considered.

Functional Requirements

The user should land on a page with tabs containing the required fields for configuration.

Login Screen

When the application starts user will be landed to a Login page with a signup link separately.

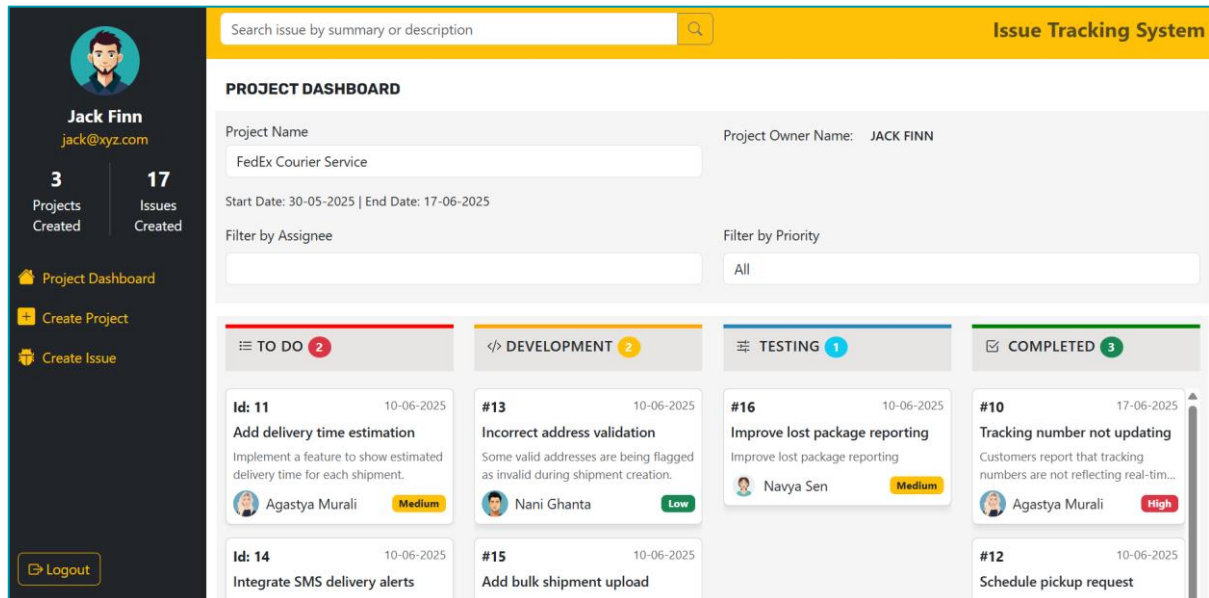
Component	Description
Issue Tracking System Login Email address Password Role: Select your role Login Don't have an account? Signup	• Email & Password fields should be required and validated for proper email and password format. • Role field should be required and should have options like 'Project Owner' and 'Assignee'. • Submit button should be disabled if form is invalid. • Signup link should navigate to signup page • Validation messages should be shown when the user interacts with the form fields. • Use Template Driven forms for this login page

Signup Screen

Clicking the 'Signup' link should navigate the user to the Signup Page, where both Assignees and Project Owners can create new accounts. Refer to the screenshot below.

Component	Description
Issue Tracking System Signup Name Email address Password Profile Role: Select your role Signup Already have an account? Login	• Name field should be mandatory and must be validated to ensure it is not left blank. • Email & Password fields should be required and validated for proper email and password format. • Profile image field should be mandatory and must be validated to ensure a valid image URL is provided in string format • Role field should be required and should have options like 'Project Owner' and 'Assignee'. • Submit button should be disabled if the form is invalid. • Login link should navigate to the login page. • Validation messages should be shown when the user interacts with the form fields. • Use Reactive forms for this signup page.

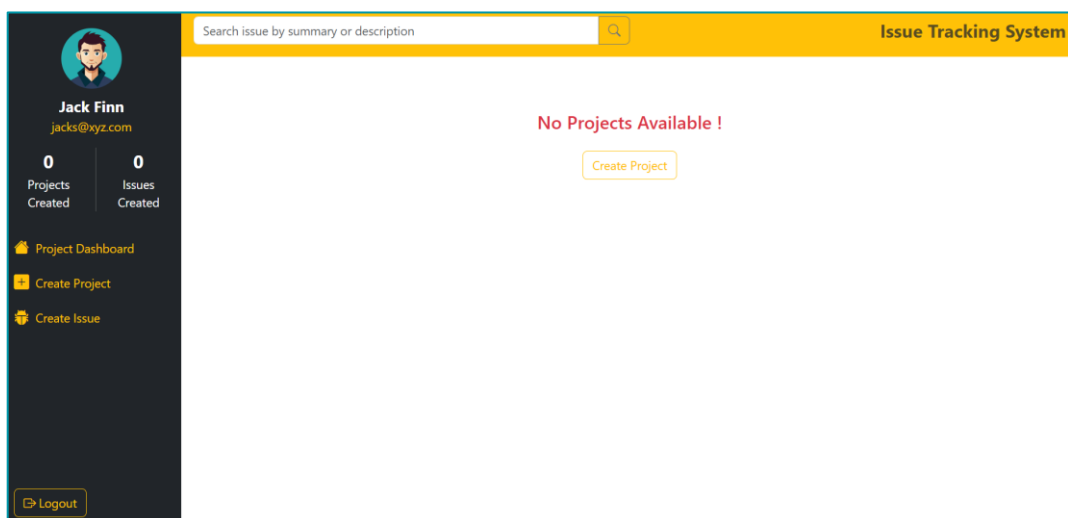
Project Owner Dashboard



The Project Owner Dashboard will have:

Components	Description
Header	Header should have Project Title in right and UI search bar on the left.
Search Bar	Search bar will filter out issue results based on summary & description if search text has a RegEx match.
Side Bar	- Should have navigation link which redirect user to access - Project dashboard, Create Issues, Create Project and a Logout button. - Should also have profile image with username, email address, number of projects created, all issues created by Project Owner logged in. - Should display content based on selection of any navigation link. By default, project dashboard will be displayed. If no projects, display “No projects Available” message & show a link below “Create Project” – which navigates to Create Project Page.

No Projects Screen



Main Content (Dashboard):

Fields	Description
Project Names	Drop-down list should be populated with project names owned by the logged-in Project Owner. The first project from the API Response should be chosen by default .
Project Owner name	Project Owner name should be displayed using a label.
Start & End Date	Start Date & End Date must be in (dd-mm-yyyy). End Date should not be lesser than Start date.
Issue Display	Issues should be displayed in card view in following categories- TO-DO, Development, Testing, Completed. The issue details should present the title, description, creation date, priority, as well as the assignee's name, and profile photo.
Filters (by assignee & priority)	Have filters as below & data filtering must be done on Front-end itself - Filter by assignee - Filter by any name of their team members. Filter by priority - Lists down issues based on selected priority.

Create Project Screen (Project Owner)



Jack Finn
jack@xyz.com

3

Projects Created

17

Issues Created

 Project Dashboard

 Create Project

 Create Issue

 Logout

Issue Tracking System

Create Project

Project Name

Enter project name

Project Owner

Project Start Date

dd-mm-yyyy

Project End Date

dd-mm-yyyy

CREATE

RESET

Component	Description
Create Form	Fields details - Project Name: Type – String, max-length (150), Required Field, No special characters except { -, / , , . }. Project Owner: Required Field, Drop-down to be fetched from API. Project Start Date & End Date: Date Picker (End date should not be lesser than start date). Validations - Reactive form with expected form validation & messages Buttons - Should have the facility to Create a Form - Only enable if form is valid. - Disable while API request is in-progress. - On Success, re-direct to project dashboard. - On Failure, show an error message from server.

- **Reset the form** - All Fields to blank.

Create Issue Screen (Project Owner)

The screenshot shows a web application interface for a Project Owner. On the left is a dark sidebar with a user profile for Jack Finn (jack@xyz.com), showing 3 Projects Created and 17 Issues Created. The main content area has a yellow header 'Deloitte Engineering' and a title 'Create Issue'. The form includes fields for Summary, Type, Project, Priority, Description, Assignee, Tags, Sprint, Story Point, and Status. At the bottom right are 'CREATE' and 'RESET' buttons.

Component	Description
Create Issues	- Form should have all the fills mentioned in above screen shot and have same look and feel. Summary: Type - String, Max Length = 150. Required Field, No special characters except { -, /, , . }. Type: Drop-down (Refer Type in Object-Value Mapping), Required. Project: Drop-down (Refer Project API values), Required. Description: Optional, Max Length = 500. Priority: Drop-down (Refer Object-Value Mapping Type), Required. Assignee: Required, Drop-down [Values from Users API]. Tags: This is a text-field, max-length = 100. Story Point: Numeric field, only accept Prime Numbers. Sprint: Numbers(1,2,3...). Validations - Reactive form with all the expected form validation should be handled – with proper validation messages and dropdown should be maintained. Buttons – Should have facility to Create a Form - Only enable if form is valid. - Disable while API request is in-progress. - On Success, re-direct to project board. - On Failure, show an error message from server. Reset the form – All Fields to blank.

Issue Detail Screen (Project Owner)

The above screen should be populated with the correct data. The following functionality is associated with this screen:

Edit Issue Details



Jack Finn
jack@xyz.com

3 Projects Created

17 Issues Created

Project Dashboard

Create Project

Create Issue

Logout

Project Board / Issue Details

Add delivery time estimation

Description :

Implement a feature to show estimated delivery time for each shipment.

Edit Issue

Status: TO_DO

Assignee: Agastya Murali

Created On: 10-06-2025

Last Updated: 10-06-2025 | 05:30

Details

Type: TASK Sprint: 2

Tags: #delivery #feature Priority: MEDIUM

Story Points: 5

Edit Issue (Project Owner)



Jack Finn
jack@xyz.com

3 Projects Created

17 Issues Created

Project Dashboard

Create Project

Create Issue

Logout

Deloitte Engineering

Edit Issue

Summary: Add delivery time estimation Type: TASK

Project: FedEx Courier Service Priority: MEDIUM

Description: Implement a feature to show estimated delivery time for each shipment.

Assignee: Agastya Murali Tags: #delivery #feature

Sprint: 2 Story Point: 5

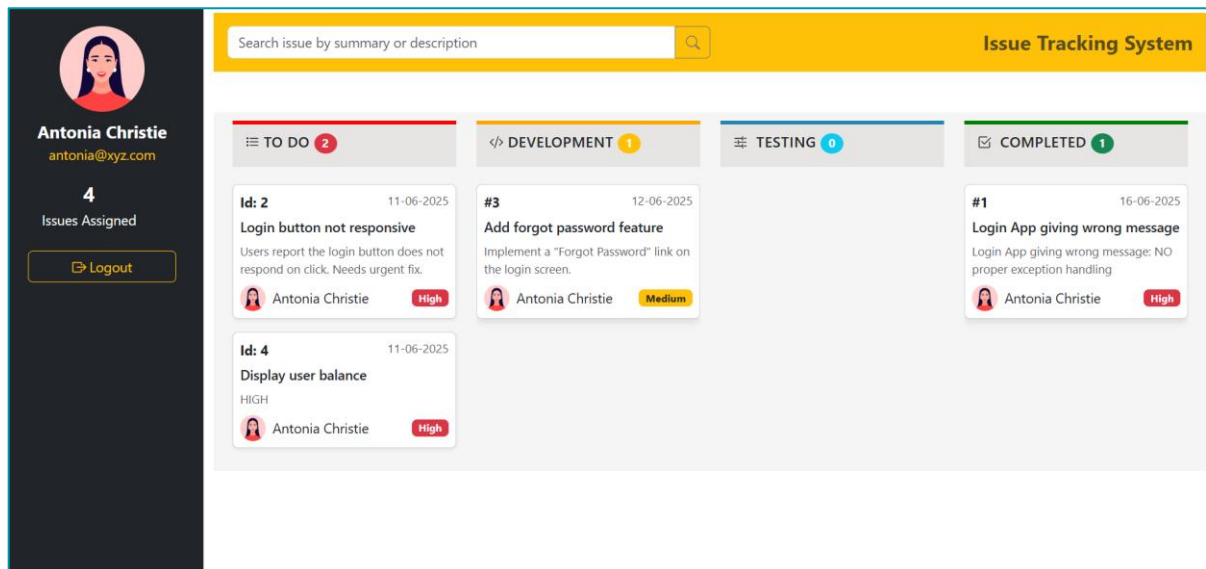
Status: TO_DO

EDIT **RESET**

Field	Description	Validation Requirements
Summary	The title of the issue.	Required, Min length: 5, Max length: 100.
Project	The project to which the issue belongs.	Required, Must be selected from a predefined list.
Type	The category of the issue.	Required, Must be selected from a predefined list (e.g., TASK, BUG, FEATURE).
Priority	The importance level of the issue.	Required, Must be selected from a predefined list (e.g., LOW, MEDIUM, HIGH).
Description	A brief explanation of the issue.	Required, Min length: 10, Max length: 500.

Assignee	The person assigned to the issue.	Required, Must be selected from a predefined list of users.
Sprint	The sprint number in which the task is planned to be completed.	Required, Must be a positive integer.
Status	The current status of the issue.	Required, Must be selected from a predefined list (e.g., TO_DO, IN_PROGRESS, DONE).
Tags	Labels or keywords associated with the issue.	Optional, Can be multiple tags, each tag should be a string.
Story Point	A numerical value indicating the effort required to complete the task.	Required, Must be a positive integer.
Edit Button	Button to submit the form and save changes.	Enabled only if all required fields are valid.
Reset Button	Button to reset the form to its initial state.	

Assignee Dashboard



Assignee Dashboard will have the following functionalities.

Components	Description
Header	Should have Application Name on right side and UI search bar on left
Search Bar	Search bar will filter out issue results based on matching RegEx between the search text & Issue summary/description.
Side Bar	Should have profile image with username, email, number of issues assigned to him/her along with Logout button to exit the dashboard
Issue Display	By default , Issues should be displayed in card view in following categories- TO-DO, Development, Testing, Completed. The issue details should present the title, description, creation date, priority, as well as the assignee's name, and profile photo.

Assignee Issue Details

Issue Dashboard / Issue Details

Login button not responsive

Description : DEVELOPMENT

Users report the login button does not respond on click. Needs urgent fix.

Details

Type: BUG Sprint: 1

Tags: #login #UI Priority: HIGH

Story Points: 3

Save Updates

- The above screen, with all mentioned respective **Issue fields** will be displayed when the user clicks on any of the **Issue card** from the Assignee Dashboard screen.
- This page features a drop-down menu for selecting the issue status.
- When the status is changed using the drop-down, the "Save Updates" button becomes enabled.
- Clicking the "Save Updates" button will update the issue's status accordingly.

Object Value Mapping

Component	Description
Status	1 => To-Do 2 => Development 3 => Testing 4 => Completed
Priority	1 => LOW 2 => MEDIUM 3 => HIGH
Type	1 => BUG 2 => TASK 3 => STORY

API Specs

List of API endpoints to be consumed:

user-controller	project-controller
GET /api/v1/users	GET /api/v1/projects/{projectId}
POST /api/v1/users	PUT /api/v1/projects/{projectId}
POST /api/v1/users/login	DELETE /api/v1/projects/{projectId}
GET /api/v1/users/{userId}	GET /api/v1/projects
GET /api/v1/users/{userId}/issues	POST /api/v1/projects
GET /api/v1/users/username/{username}/issues	GET /api/v1/projects/{projectId}/issues
	GET /api/v1/projects/{projectId}/insights
	GET /api/v1/projects/projectName/{projectName}/issues
	GET /api/v1/projects/owner/{ownerId}

issue-controller	
GET	/api/v1/issues/{id}
PUT	/api/v1/issues/{id}
GET	/api/v1/issues
POST	/api/v1/issues
GET	/api/v1/issues/project/{projectId}
GET	/api/v1/issues/owner/{ownerId}
GET	/api/v1/issues/assignee/{assigneeId}

Non-Functional Requirements

- The application should have low latency and high throughput.
- The application and data should be secured by Authentication.
- System should be scalable and maintainable.

Guidelines

- Make use of best coding practices.
- Develop the application so that it facilitates the easy plugging of other related modules like getting real-time updates and targeted notifications in case of status changes, in-issue commenting, etc. while scaling the application later.
- You can make assumptions and incorporate additional features or functionalities.
- A simple but working prototype is preferred over a non-working but good-looking application.
- The screenshots provided in this document are intended to help you understand and can be used as a reference to build the application.
- Add relevant comments for proper documentation.
- There would be a heavy focus on code quality and robustness of code in general.
- Use Git to maintain project repository, manage pull requests, and resolve conflicts.

Best Practices

- Follow consistent React coding styles with proper naming conventions.
- Ensure use of heavy type scripting. Avoid using 'any'.
- Use ES6 features.
- Build reusable components.
- Add comments for each variable and method.
- Project structure and folder structure must be maintained.

Milestones and Evaluation Criteria

S. No.	Details
Milestone 1	Login and Sign-Up - Implementation Functionality: Refer 4.1 & 4.2
Milestone 2	Project Owner Dashboard - Implementation Functionality: Refer 4.3
Milestone 3	Create project - Implementation Functionality: Refer 4.4
Milestone 4	Create Issue - Implementation Functionality: Refer 4.5
Milestone 5	Issue Details - Implementation Functionality: Refer 4.6
Milestone 6	Assignee Dashboard - Implementation Functionality: Refer 4.8
Milestone 7	Back End API Integration- Create the needed Services, Models and Integrate all the above Screens implementation with Back-end API developed as a part of Week 2 Case Study
Milestone 8	Push the build to your GitHub repository
Technical Interview	Discussion of Assumptions, Functionalities, and Validations



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